

Statistics of Financial Markets

Seminar 13: Machine learning



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **GB**: Gradient Boosting
- **HAR**: Heterogeneous AutoRegressive (model)
- **HS**: Historical Simulation
- **ML**: Machine Learning
- **MLP**: MultiLayer Perceptron
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **QGB**: Quantile Gradient Boosting
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QR**: Quantile Regression
- **RF**: Random Forest
- **ROC**: Receiver Operating Characteristic (curve)
- **SPX**: S&P 500 index (ticker symbol)
- **SSE**: Sum of Squared Errors
- **VIX**: CBOE Volatility IndeX

Today's Question and Seminar Structure

- Question of the seminar
  - **when does a flexible prediction method beat a simple model on financial data, and how do we test it without fooling ourselves?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the Exercise Map (next slide)
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	bias, variance and expected error from five training sets	Solved, Proposed	A1
A3, A4	the best split of a regression tree, step by step	Solved, Proposed	A3
A5, A6	ridge and lasso with one feature	Solved, Proposed	A5
A7, A8	a walk-forward design; a purged K-fold with an embargo	Solved, Proposed	A7
B1, B2	volatility forecasts: S&P 500 (next day), DAX (next week)	Solved, Proposed	B1
B3, B4	the sign of daily returns: DAX; TLV and SNP	Solved, Proposed	B3
B5, B6	VaR 1% of the DAX by quantile regression and quantile boosting	Solved, Proposed	B5
C1, C2	does the S&P 500 help to forecast BET volatility? what is wrong in an AI answer?	Proposed	B2, A7

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model in the last column
- Part A: calculations on paper; Part B: real data, in the notebook, each ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data Used

Series	Source	Prices used	Period
S&P 500	EODHD	open, high, low, close	2008–2026
DAX	EODHD	open, high, low, close	2006–2026
BET	EODHD	close	2008–2026
Banca Transilvania (TLV), OMV Petrom (SNP)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; daily variance proxy v_t from open, high, low and close (Chapter 8); the last day is 18 September 2026
- In the notebook: `rv_frame('dax')`, `walk_forward(...)`, `rv_metrics(...)`, `sign_walk_forward('dax')`, `var_backtest(...)` (scikit-learn); no account or key is needed

Prerequisites for Today (1/20): Learning from Data

- **Machine learning** (ML): choose from data a prediction rule $\hat{f}$ with a small error on **new** data
  - x_t: the **features**, the numbers known at the close of day t (past volatilities, past returns)
  - y_t: the **target**, the quantity to forecast; $\hat{y}_t = \hat{f}(x_t)$: the forecast (a hat marks an estimate)
- **Loss** and **mean squared error** (MSE) on n days:

$$\mathrm{MSE} = \frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2$$

- **Training error**: the MSE on the days used to fit $\hat{f}$; **test error**: the MSE on days not used
  - **overfitting**: a small training error and a large test error: the model has learned the noise
  - a **hyperparameter** (tree depth, penalty λ) sets the flexibility; it is chosen on data not used for the fit

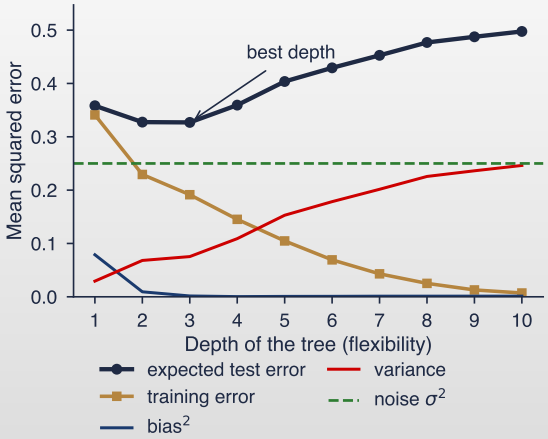
Prerequisites for Today (2/20): Bias and Variance

- Model: $y = f(x) + \varepsilon$, with $E[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$; $\hat{f}$ is fitted on a random training sample
  - f: the true, unknown relation; ε: the noise; $E[\cdot]$: the mean over training samples and noise
- Expected squared error at a point x_0, with $y_0 = f(x_0) + \varepsilon_0$:

$$E[(y_0 - \hat{f}(x_0))^2] = \underbrace{(E\hat{f}(x_0) - f(x_0))^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{noise}}$$

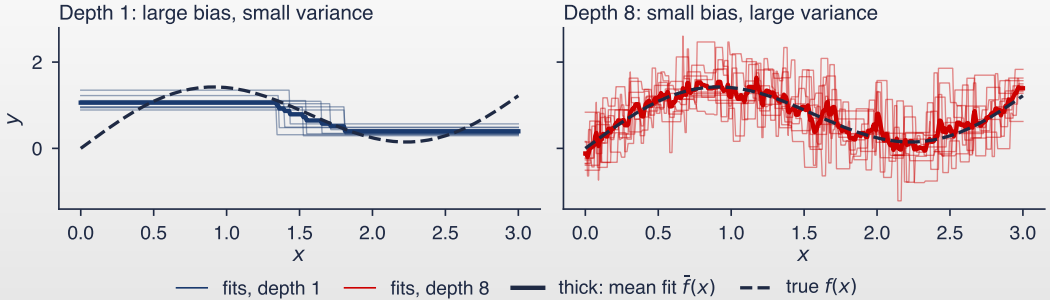
- From M training sets with forecasts $\hat{f}_1, \ldots, \hat{f}_M$ at x_0:
  - $\bar{f} = \frac{1}{M}\sum_{m=1}^{M} \hat{f}_m$: the mean forecast; bias $= \bar{f} - f(x_0)$: the systematic error
  - variance $= \frac{1}{M}\sum_{m=1}^{M}(\hat{f}_m - \bar{f})^2$: how much the forecast changes from one sample to another

Prerequisites for Today (3/20): Training Error and Test Error



- Simulation of the model of (2/20): trees of depth 1 to 10 fitted on 200 training samples
  - depth: the number of successive splits, the flexibility of the tree
- **Training error** (ochre) falls towards 0 as the depth grows
  - a deep tree fits every training point, including the noise
- **Expected test error** (navy), the sum of the three terms of (2/20): U-shaped
  - bias^2 (blue) falls, the variance (red) rises; the noise σ^2 (green) cannot be removed
  - the best depth is where the test error is smallest, not where the training error is

Prerequisites for Today (4/20): Shallow and Deep Trees on Ten Samples



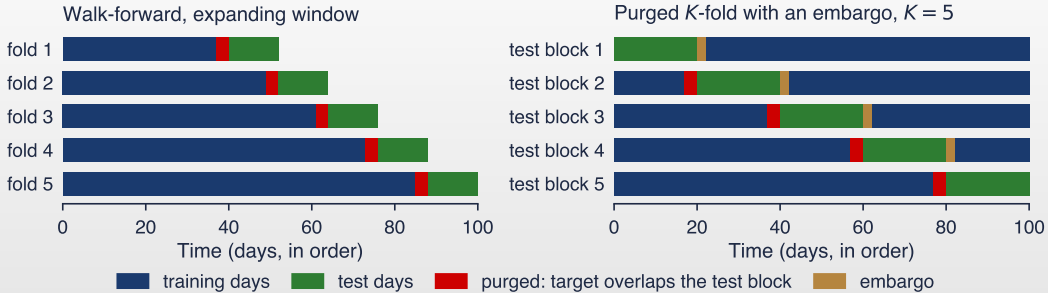
- Thin lines: the fits on ten simulated training samples; thick line: their mean $\bar{f}(x)$; dashed: the true $f(x)$
  - depth 1 (left): the ten fits almost coincide (small variance), but $\bar{f}$ misses the shape of f (large bias)
  - depth 8 (right): $\bar{f}$ follows f (small bias), but each fit jumps with the noise of its sample (large variance)

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Prerequisites for Today (5/20): Validation for Time Series

- **Walk-forward**: train on the past, test on the next block (here: one calendar year), refit, move forward
  - expanding window: the training sample keeps all earlier days
- **Purging**: the target y_t uses the days $t+1, \ldots, t+h$
  - h: the horizon of the target, in days; the last h training days have targets inside the test block, so they are dropped
  - **K-fold**: K blocks in time order, each the test block once; **embargo**: also drop a few training days right after the test block
- **Look-ahead bias**: a feature uses information not yet known at time t (future prices, full-sample means)
  - **leakage**: information about test targets reaches training
  - **random K-fold** (shuffled folds) causes it with overlapping targets

Prerequisites for Today (6/20): Walk-Forward and Purged K-Fold



- Left: each fold trains on all earlier days (blue), drops the last h of them (red) and tests on the next block (green)
- Right: K-fold in time order; training days on both sides of the test block
  - before the test block: purge h days; after it: an embargo (ochre); training days = all − test − purged − embargo

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Prerequisites for Today (7/20): Ridge and Lasso

- **Ridge**: minimise the squared error plus a penalty on the squared coefficients

$$\hat{\beta}^R = \arg\min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j \beta_j^2$$

- Notation
  - $x_t^\top \beta = \sum_j \beta_j x_{tj}$: the linear forecast; β_j: the coefficient of feature j; arg min: the minimising value
  - $\lambda \geq 0$: the penalty; $\lambda = 0$ gives OLS (ordinary least squares), a large λ pushes the coefficients towards 0
- **Lasso**: the same criterion with the absolute values of the coefficients

$$\hat{\beta}^L = \arg\min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j |\beta_j|$$

- Reading
  - the lasso sets some coefficients exactly to zero: it selects features
  - both trade a little bias for a smaller variance; standardise the features with the training mean and standard deviation first

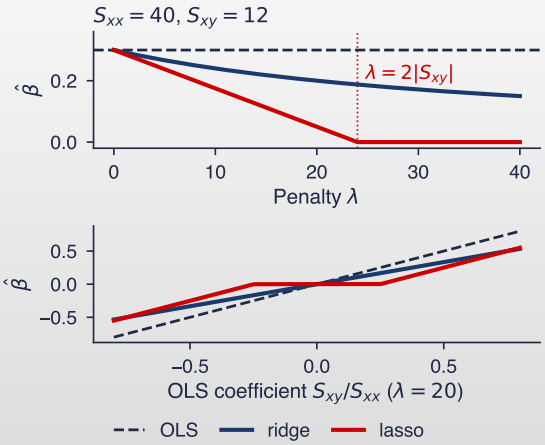
Prerequisites for Today (8/20): Ridge and Lasso with One Feature

- **One centred feature** (mean 0), no intercept
  - $S_{xx} = \sum_t x_t^2$: the sum of squares of x; $S_{xy} = \sum_t x_t y_t$: the sum of cross products
- The three estimators:

$$\hat{\beta}^{OLS} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}^{R} = \frac{S_{xy}}{S_{xx} + \lambda}, \qquad \hat{\beta}^{L} = \frac{\operatorname{sign}(S_{xy}) \max(|S_{xy}| - \lambda/2,\ 0)}{S_{xx}}$$

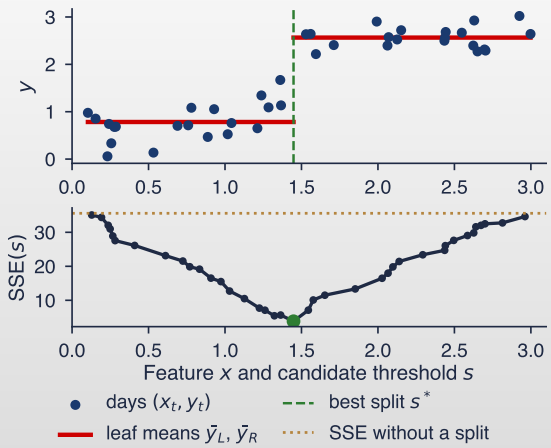
- Ridge
  - the OLS coefficient times the **shrinkage factor** $S_{xx}/(S_{xx} + \lambda) < 1$; it tends to 0 but never reaches it
- Lasso (**soft thresholding**)
  - $\operatorname{sign}(S_{xy})$: $+1$ or -1; $\max(a, 0)$: a if positive, 0 otherwise
  - the lasso subtracts $\lambda/2$ from $|S_{xy}|$; if $|S_{xy}| \leq \lambda/2$, the coefficient is exactly 0

Prerequisites for Today (9/20): Shrinking and Selecting



- Top: the coefficient as a function of λ, for $S_{xx} = 40$, $S_{xy} = 12$
  - OLS: $12/40 = 0.30$ for every λ
  - ridge (blue): a smooth decline, always above 0
  - lasso (red): a straight decline, exactly 0 from $\lambda = 2|S_{xy}| = 24$
- Bottom: the estimate against the OLS coefficient, for a fixed $\lambda = 20$
  - ridge multiplies every coefficient by the same factor $40/60$
  - lasso subtracts $\lambda/(2S_{xx}) = 0.25$ and sets the small coefficients to 0

Prerequisites for Today (10/20): A Regression Tree



- **Split**: the days with $x_j \leq s$ go left, the others right; each leaf predicts the mean of its y
  - j: the feature used; s: the threshold
- Choose (j, s) with the smallest sum of squared errors (SSE):

$$\mathrm{SSE}(s) = \sum_{x_{tj} \leq s} (y_t - \bar{y}_L)^2 + \sum_{x_{tj} > s} (y_t - \bar{y}_R)^2$$

- Notation and reading
  - $\bar{y}_L$, $\bar{y}_R$: the means of y in the left and the right leaf
  - candidates for s: the midpoints between consecutive sorted values of x_j
  - share of the SSE removed: $1 - \mathrm{SSE}(s^*)/\mathrm{SSE}_0$, with $\mathrm{SSE}_0 = \sum_t (y_t - \bar{y})^2$

Prerequisites for Today (11/20): Random Forest and Gradient Boosting

- **Random forest** (RF): the average of B deep trees

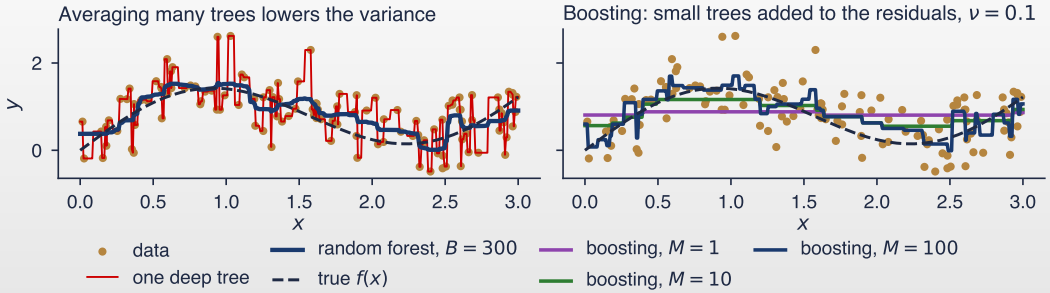
$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=\mathbf{1}}^{B} \hat{T}_b(x)$$

- Notation and reading
  - $\hat{T}_b$: tree b, grown on a **bootstrap** sample (days drawn at random with replacement), with a random subset of features at each split
  - averaging B weakly correlated trees lowers the variance; the bias stays that of one deep tree (small)
- **Gradient boosting** (GB): small trees added one by one, each fitted to the current residuals

$$F_m(x) = F_{m-\mathbf{1}}(x) + \nu \, \hat{T}_m(x), \qquad \hat{T}_m \text{ fitted to } y_t - F_{m-\mathbf{1}}(x_t)$$

- Notation and reading
  - $F_{\mathbf{0}}$: the mean of y; F_m: the model after m trees; M: the number of trees
  - $\nu \in (0, 1]$: the **learning rate**; a small ν needs more trees but overfits less

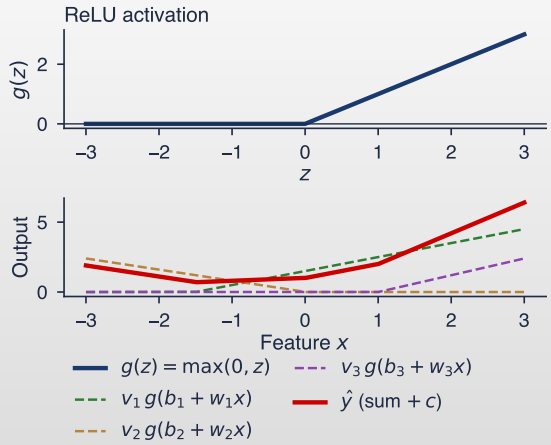
Prerequisites for Today (12/20): Averaging Trees and Boosting



- Left: one deep tree (red) follows every simulated point; the forest of 300 trees (blue) is smooth and close to f
- Right: with $\nu = 0.1$, boosting after $M = 1$ tree is almost the mean; after 10 trees it has the broad shape; after 100 it fits the detail

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Prerequisites for Today (13/20): Neural Networks



- **MLP** (multilayer perceptron) with one hidden layer of K neurons:

$$\hat{y} = c + \sum_{k=1}^{K} v_k \, g(b_k + w_k^\top x)$$

- Notation
  - w_k, b_k: the weights and the constant of neuron k
  - v_k, c: the weights and the constant of the output
  - $g(z) = \max(0, z)$ (ReLU): the activation function (top)
- Reading (bottom)
  - each neuron adds a kink at $x = -b_k / w_k$
  - the sum of K neurons is a piecewise-linear function: a nonlinear forecast
  - the weights are fitted by minimising the MSE on the training days

Prerequisites for Today (14/20): The Daily Variance Proxy

- **Daily variance proxy** v_t (Chapter 8), from the open O_t, high H_t, low L_t and close C_t:

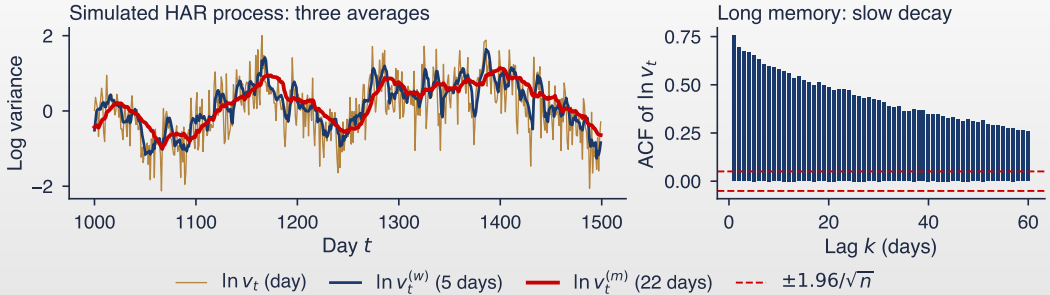
$$v_t = o_t^2 + \tfrac{1}{2}(u_t - d_t)^2 - (2\ln 2 - 1)\, c_t^2$$

- Notation (log prices in %)
  - $o_t = 100 \ln(O_t / C_{t-1})$: the overnight move; $u_t = 100 \ln(H_t / O_t)$, $d_t = 100 \ln(L_t / O_t)$: high and low against the open
  - $c_t = 100 \ln(C_t / O_t)$; the last two terms are the Garman–Klass estimator of the variance during the session
- **Target** for horizon h (B1: $h = 1$; B2, C1: $h = 5$) and **variance forecast**:

$$y_t = \ln\Big(\frac{1}{h}\sum_{j=1}^{h} v_{t+j}\Big), \qquad \hat{h}_t = \exp(\hat{y}_t + s^2/2)$$

- Reading
  - logs make the target close to Normal; s^2: the variance of the training residuals of y, which corrects the bias of $\exp(\hat{y}_t)$

Prerequisites for Today (15/20): The HAR Benchmark



- **HAR** (heterogeneous autoregressive) model, estimated by OLS: $y_t = \beta_0 + \beta_d \ln v_t + \beta_w \ln v_t^{(w)} + \beta_m \ln v_t^{(m)} + u_t$
- $v_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} v_{t-j}$, $v_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} v_{t-j}$: the means over the last week and month; u_t: the error
  - β_d, β_w, β_m: the weights of the day, the week and the month; three averages reproduce the slow decay of the ACF (autocorrelation function, right)

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Prerequisites for Today (16/20): Evaluating Forecasts

- **Out-of-sample** R^2 against a benchmark forecast $\tilde{y}_t$ (the training mean or HAR):

$$R^2_{OOS} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \tilde{y}_t)^2}$$

- Reading
  - the sums run over the test days only; $R^2_{OOS} > 0$: better than the benchmark; < 0: worse; reported in %
- **QLIKE loss** of a variance forecast $\hat{h}_t$ for the realised variance RV_t:

$$L_t = \frac{\mathrm{RV}_t}{\hat{h}_t} + \ln \hat{h}_t$$

- Reading
  - $\mathrm{RV}_t = \frac{1}{h} \sum_{j=1}^{h} v_{t+j}$: the variance that followed; L_t is smallest at $\hat{h}_t = \mathrm{RV}_t$
  - compare models by the mean QLIKE over the test days: lower is better

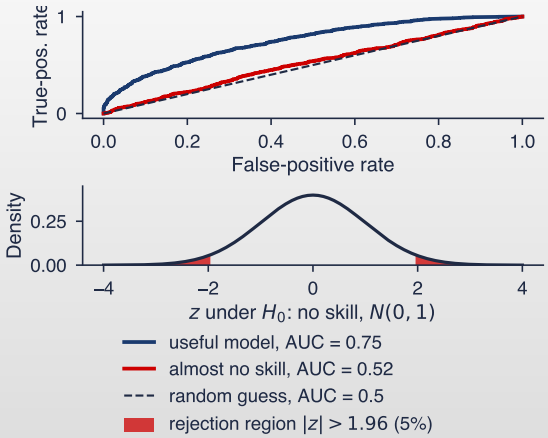
Prerequisites for Today (17/20): The Diebold–Mariano Test

- **DM** (Diebold–Mariano) test of equal accuracy of models A and B on n test days
  - loss difference $d_t = L_t^A - L_t^B$; H_0: $E[d_t] = 0$ (equal accuracy)
- Statistic, with a Newey–West long-run variance of d_t:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_{LR}^2/n}}, \qquad \hat{\sigma}_{LR}^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell}\Big(1 - \frac{j}{\ell+1}\Big)\hat{\gamma}_j$$

- Notation
  - $\bar{d}$: the mean loss difference; $\hat{\gamma}_j$: the sample autocovariance of d_t at lag j
  - ℓ: the number of lags, here the horizon h: overlapping targets make d_t autocorrelated
- Reference distribution and reading
  - under H_0, DM $\approx N(0,1)$; $|\mathrm{DM}| > 1.96$: the accuracies differ at 5%
  - DM < 0: model A has the smaller loss; DM > 0: model B

Prerequisites for Today (18/20): Classification



- **Logit**: probability that tomorrow is an up day

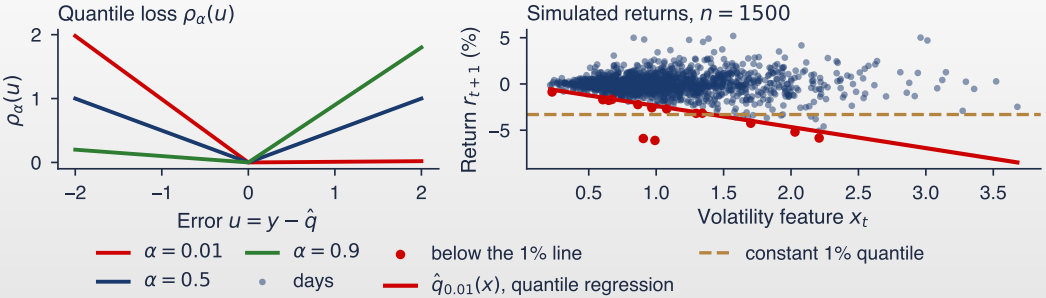
$$\hat{p}_t = \frac{1}{1 + e^{-(\beta_0 + x_t^\top \beta)}}$$

- forecast "up" if $\hat{p}_t \geq 0.5$; accuracy: the share of correct signs
  - baseline acc_0: the accuracy of always forecasting the more frequent class
- Test of skill against the baseline:

$$z = \frac{\widehat{\mathrm{acc}} - \mathrm{acc}_0}{\sqrt{\mathrm{acc}_0(1 - \mathrm{acc}_0)/n}}$$

- n: the test days; $z \approx N(0, 1)$ without skill (bottom); $z > 1.96$: significantly better
  - **AUC**: the area under the ROC curve (top) = the probability that a random up day gets a higher $\hat{p}$ than a random down day; 0.5: no skill

Prerequisites for Today (19/20): Quantile Regression



- $\hat{q}_\alpha(x) = x^\top \hat{\beta}_\alpha$, the conditional α-quantile, minimises $\sum_t \rho_\alpha(y_t - x_t^\top \beta)$, with $\rho_\alpha(u) = u\,(\alpha - 1\{u < 0\})$ (left)
- $1\{u < 0\}$: 1 if $u < 0$, 0 otherwise; an error below the quantile costs $(1-\alpha)|u|$, above it $\alpha|u|$
  - right: the 1% line falls with volatility, so the VaR rises on volatile days; VaR 1% $= -\hat{q}_{0.01}$

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Prerequisites for Today (20/20): VaR by Historical Simulation and Backtesting

- **Historical simulation** (HS): VaR 1% $= -\hat{q}_{\mathbf{0.01}}$ of the last 500 returns (the empirical quantile)
  - **exception**: a day with $r_{t+\mathbf{1}} < -\mathrm{VaR}_t$; x: the number of exceptions in n days; $\hat{\pi} = x/n$
- **Kupiec** test of the rate, $H_{\mathbf{0}}$: $\pi = \alpha$:

$$LR_{uc} = -2\big[(n-x)\ln(1-\alpha) + x\ln\alpha - (n-x)\ln(1-\hat{\pi}) - x\ln\hat{\pi}\big] \sim \chi^{\mathbf{2}}(1)$$

- **Christoffersen** test of independence, $H_{\mathbf{0}}$: an exception today does not change the chance of one tomorrow:

$$LR_{ind} = -2\big[\ln L(\hat{\pi}) - \ln L(\hat{\pi}_{\mathbf{01}}, \hat{\pi}_{\mathbf{11}})\big] \sim \chi^{\mathbf{2}}(1)$$

- Notation and reading
  - $\hat{\pi}_{\mathbf{01}}$, $\hat{\pi}_{\mathbf{11}}$: the shares of exceptions after a day without and with an exception; L: the likelihood of the 0–1 sequence
  - reject at 5% if $LR > 3.84$; average quantile loss $\frac{\mathbf{1}}{n}\sum_t \rho_\alpha(r_{t+\mathbf{1}} + \mathrm{VaR}_t)$: lower is better

A1: Bias and Variance of a Shallow Tree [Solved]

Task

- At a point x_0 the true value is $f(x_0) = 2.0$ and $\sigma^2 = 0.25$. A depth-1 tree fitted on five training sets forecasts 1.6, 1.7, 1.5, 1.8, 1.6.
- 1. Compute the mean forecast and the bias.
- 2. Compute the variance of the five forecasts (divide by 5).
- 3. Compute the expected squared error at x_0.
- Report: four numbers and one sentence.

Solution

- 1. $\bar{f} = 8.2/5 = 1.640$; bias $= 1.640 - 2 = -0.360$
- 2. $[(-0.04)^2 + 0.06^2 + (-0.14)^2 + 0.16^2 + (-0.04)^2]/5 = 0.0104$
- 3. $0.1296 + 0.0104 + 0.25 = 0.390$
- The shallow tree is stable but systematically too low: its error is mostly squared bias.

Back Exercise list

A2: Bias and Variance of a Deep Tree [Proposed]

Solution discussed in class

Task

- Same point and noise as in A1. A depth-10 tree on the same five training sets forecasts 2.6, 1.4, 2.3, 1.7, 2.2. Model: A1.
- 1. Compute the mean forecast, the bias and the variance.
- 2. Compute the expected squared error.
- 3. Decide which tree you would use at x_0.
- Report: four numbers and one sentence.

A3: One Split of a Regression Tree [Solved]

Task

- Eight days: x = yesterday's absolute return (%), y = today's variance proxy:
- x: 0.2, 0.4, 0.5, 0.9, 1.1, 1.6, 2.0, 2.8; y: 0.5, 0.7, 0.6, 1.0, 1.4, 2.2, 2.6, 3.4
- 1. Compute the mean of y and the SSE without a split.
- 2. For each candidate split, compute both leaf means and the total SSE.
- 3. Choose the best split and the share of the SSE it removes.
- Report: the table, the split and one sentence.

Solution

s	$\bar{y}_L$	$\bar{y}_R$	SSE
0.30	0.500	1.700	6.740
0.45	0.600	1.867	5.593
0.70	0.600	2.120	3.668
1.00	0.700	2.400	2.220
1.35	0.840	2.733	1.279
1.80	1.067	3.000	2.393
2.40	1.286	3.400	4.089

- $\bar{y} = 1.550$, SSE $= 8.000$; best $s = 1.35$: leaves 0.840 and 2.733, SSE $= 0.532 + 0.747 = 1.279$
- the split removes $1 - 1.279/8.000 = 0.840$ of the SSE: large moves yesterday, high variance today

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A4: A Split on the Volatility Index [Proposed]

Solution discussed in class

Task

- Eight months: x = the VIX level (the implied volatility index of the S&P 500, annual %) at the end of the month, y = next month's realised volatility (% per day). Model: A3.
- x: 12, 14, 15, 17, 21, 24, 30, 38; y: 0.8, 0.9, 0.7, 1.1, 1.5, 1.4, 2.6, 3.0
- 1. Compute the SSE without a split and for every candidate split.
- 2. Choose the best split and give both leaf forecasts.
- 3. Compute the share of the SSE that the split removes.
- Report: the table, three numbers and one sentence.

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A5: Ridge with One Feature [Solved]

Task

- One centred feature with $S_{xx} = 50$ and $S_{xy} = 20$; no intercept.
- 1. Compute the OLS coefficient.
- 2. Compute the ridge coefficient and the shrinkage factor $S_{xx}/(S_{xx} + \lambda)$ for $\lambda = 10$ and $\lambda = 50$.
- 3. Explain what happens when $\lambda \to \infty$.
- Report: five numbers and one sentence.

Solution

- 1. $\hat{\beta}^{OLS} = 20/50 = 0.40$
- 2. $\lambda = 10$: $20/60 = 0.333$, factor 0.833; $\lambda = 50$: $20/100 = 0.20$, factor 0.50
- 3. The coefficient tends to 0 but never equals 0.
- Ridge multiplies the OLS coefficient by a factor below 1: it shrinks, it does not select.

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A6: Lasso with One Feature [Proposed]

Solution discussed in class

Task

- Same data as in A5: $S_{xx} = 50$, $S_{xy} = 20$. Model: A5.
- 1. Compute the lasso coefficient for $\lambda = 10$, 30 and 50.
- 2. Find the smallest λ for which the coefficient is exactly zero.
- 3. Compare the lasso with the ridge results of A5 for $\lambda = 50$.
- Report: four numbers and one sentence.

A7: A Walk-Forward Design [Solved]

Task

- Daily S&P 500 features from 8 April 2008 to 18 September 2026; target: the mean variance of the next 5 days; refits each January from 2013.
- 1. Count the folds of the walk-forward evaluation.
- 2. For the 2013 fold, say which training days must be purged and why.
- 3. Explain why no embargo is needed.
- 4. Explain what would go wrong if the 2013 model were tuned by random 5-fold on 2008–2012.
- Report: two numbers and three sentences.

Solution

- 1. One fold per year 2013–2026: 14 folds.
- 2. The last 5 training days (from 24 December 2012): their targets use January 2013; the fold trains on 1 188 days, the last one 21 December 2012, and tests on 252.
- 3. All training days come before the test year: no training feature uses test information.
- 4. Random folds put neighbouring days with overlapping targets in training and validation: the chosen hyperparameters reward memorising, not forecasting.

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A8: A Purged K-Fold with an Embargo [Proposed]

Solution discussed in class

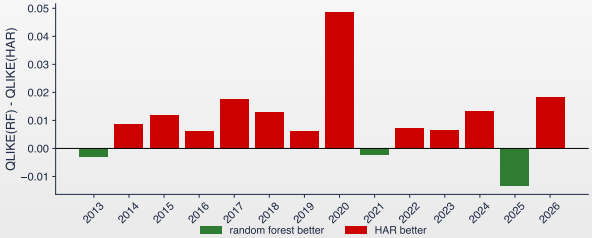
Task

- 2500 days, $K = 5$ blocks in time order, a target that uses the next 21 days, an embargo of 1% of the days. Model: A7.
- 1. Compute the size of each test block and of the embargo.
- 2. Compute the number of training days when the test block is in the middle.
- 3. Compute it when the test block is the first and when it is the last block.
- Report: four numbers and one sentence.

B1: Next-Day Volatility of the S&P 500 [Solved]

- **Question**: does a random forest forecast tomorrow's variance of the S&P 500 better than HAR?
- **Inputs**: S&P 500 open, high, low and close since 2008; target: $\ln v_{t+1}$; features: the three HAR averages
- **Tasks**
  1. Fit HAR (OLS) and a random forest (300 trees, at least 20 days per leaf) walk-forward, refitting each January from 2013.
  2. Compute the out-of-sample R^2 of both against the training mean and of the forest against HAR.
  3. Compute the mean QLIKE of both and the Diebold–Mariano test of equal QLIKE.
  4. Draw the QLIKE difference per year.
  5. Interpretation: is the forest worth its complexity here?
- **Report**: four numbers, a test with its p-value, the chart and two sentences
- Notebook: section B1

B1: Solution [Solved]



- 3 448 days from 2 January 2013; R^2_{OOS} against the mean: HAR 54.8%, forest 53.4%; forest against HAR: -3.2%
- QLIKE: HAR 0.329, forest 0.339; DM $= +2.65$ (p 0.008); the forest is better in only 3 of 14 years
- Interpretation: no: with the same three features the forest is significantly worse than the linear HAR; the relation is close to linear in logs and the forest adds variance

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B2: Weekly Volatility of the DAX [Proposed]

- **Question**: do lasso, random forest, gradient boosting or an MLP beat HAR for next week's DAX variance? Model: B1.
- **Inputs**: DAX open, high, low and close since 2006; target: ln of the mean v over the next 5 days; 11 features (HAR, 4 lags, the 66-day mean, r_t, $\min(r_t, 0)$, $|r_t|$)
- **Tasks**
  1. Fit the five models walk-forward from 2012, purging the last 5 training days.
  2. Compute the out-of-sample R^2 against HAR and the mean QLIKE of each model.
  3. Run the Diebold–Mariano test of each model against HAR.
  4. Interpretation: which model would you use for the DAX?
- **Report**: one table and two sentences
- Notebook: section B2

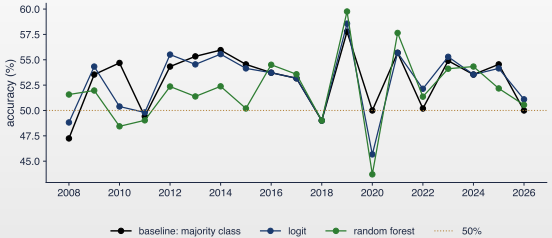
Back Exercise list

B3: The Sign of DAX Returns [Solved]

- **Question**: can a logit or a random forest predict whether the DAX rises tomorrow better than "always up"?
- **Inputs**: DAX closes since 2000; target $1\{r_{t+1} > 0\}$; features: the last 5 returns, sums over 5, 21 and 63 days, volatility over 21 and 63 days
- **Tasks**
  1. Fit both classifiers walk-forward from 2008, refitting each January.
  2. Compute the accuracy of each model and of the majority-class baseline.
  3. Test each accuracy against the baseline with the z statistic, and compute the AUC.
  4. Draw the accuracy by year.
  5. Interpretation: does either model have skill?
- **Report**: six numbers, the chart and two sentences
- Notebook: section B3

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B3: Solution [Solved]



- $n = 4\,752$ days; up days 53.1% = the baseline accuracy; logit 52.9% ($z = -0.20$, p 0.839, AUC 0.507); forest 52.0% ($z = -1.45$, p 0.146, AUC 0.508)
- The logit forecasts "up" on 90% of the days; the forest beats the baseline in 8 of 19 years, as often as chance would allow
- Interpretation: no: both accuracies are at or below the baseline, and the AUC is close to 0.5; daily DAX returns are close to unpredictable (Chapter 7)

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B4: The Sign of Two BVB Stocks [Proposed]

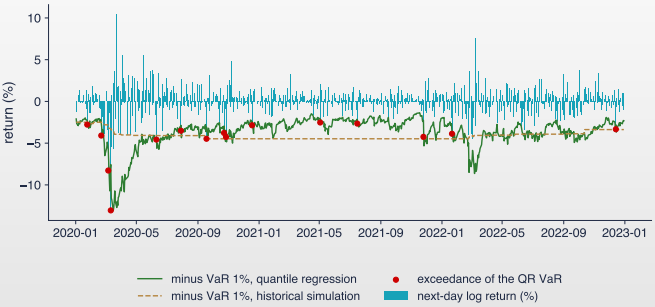
- **Question**: is the sign of Banca Transilvania and OMV Petrom returns easier to predict than that of the DAX? Model: B3.
- **Inputs**: TLV (without 30–31 May 2016) and SNP adjusted closes since 2010; the features of B3; forecasts from 2014
- **Tasks**
  1. Fit the logit, random forest and gradient boosting walk-forward from 2014.
  2. Compute the accuracy, the baseline accuracy, the z statistic and the AUC for each stock and model.
  3. Interpretation: is there more predictability on the BVB than on the DAX?
- **Report**: one table and two sentences
- Notebook: section B4

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B5: VaR 1% of the DAX by Quantile Regression [Solved]

- **Question**: does a linear quantile regression on volatility features give a better VaR 1% than historical simulation?
- **Inputs**: DAX since 2006; target $r_{t+\mathbf{1}}$; features: $\sqrt{v_t}$, $\sqrt{v^{(w)}}$ and $\sqrt{v^{(m)}}$ (square roots of the mean v over 5 and 22 days), r_t
- **Tasks**
  1. Fit the 1% quantile regression walk-forward from 2012 and set VaR 1% $= -\hat{q}_{\mathbf{0.01}}$.
  2. Compute the historical-simulation VaR 1% on the last 500 days.
  3. Count the exceptions and run the Kupiec and Christoffersen tests; compute the average quantile loss.
  4. Draw both VaR forecasts in 2020–2022.
  5. Interpretation: which VaR would you report to a risk committee?
- **Report**: a table, the chart and two sentences
- Notebook: section B5

B5: Solution (1/2) [Solved]



- 3 731 days from 2 January 2012; last fit: $\hat{q}_{\mathbf{0.01}} = -0.472 - 0.434\sqrt{v_t} - 1.814\sqrt{v^{(w)}} - 0.221\sqrt{v^{(m)}} + 0.158 r_t$; VaR 1% on 17 September 2026: 2.24%

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B5: Solution (2/2) [Solved]

Method	x	rate (%)	p Kupiec	p ind.	loss $\times 100$
QR	39	1.05	0.783	0.364	3.80
HS	49	1.31	0.066	< 0.001	4.75

- Columns
  - x: the number of exceptions; rate $= x/n$, target 1%; loss $\times 100$: the mean quantile loss
  - p Kupiec, p ind.: the p-values of the Kupiec (rate) and Christoffersen (independence) tests
- Interpretation: the quantile regression: right rate, independent exceptions, smaller loss; HS is exceeded in clusters (13 times in 2020) because a 500-day window reacts slowly

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B6: Quantile Gradient Boosting for the DAX [Proposed]

- **Question**: does a flexible quantile model improve on the linear quantile regression of B5? Model: B5.
- **Inputs**: as in B5; gradient boosting with the quantile loss at 1%, 300 trees of depth 2, learning rate 0.05, at least 50 days per leaf
- **Tasks**
  1. Fit the quantile boosting walk-forward from 2012 and compute its VaR 1%.
  2. Count the exceptions, run the Kupiec and Christoffersen tests and compute the quantile loss.
  3. Interpretation: is the flexible model worth it at the 1% level?
- **Report**: one table and two sentences
- Notebook: section B6

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C1: Does the S&P 500 Help to Forecast BET Volatility? [Proposed]

- **Question**: the BVB closes before New York; does yesterday's S&P 500 volatility improve a HAR forecast of next week's BET variance?
- **Inputs**: BET and S&P 500 closes since 2008; proxy: squared daily returns (the BET has no high and low prices in our data); models: B1, B2
- **Tasks**
  1. Build the HAR features of the BET and the HAR features of the S&P 500 at the previous New York close.
  2. Compare HAR, HAR plus the S&P 500 features (OLS) and a random forest on all features, walk-forward from 2012.
  3. Compute the out-of-sample R^2, QLIKE and the Diebold–Mariano tests against HAR.
  4. Interpretation: is the S&P 500 information worth adding?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise machine learning for the S&P 500. The answer:
- (a) With random 5-fold cross-validation a random forest explains 36.9% of the next 21-day return: the market is predictable.
- (b) The forest fits next week's log variance better than HAR: in-sample R2 80.2% against 73.7%.
- (c) A forest predicts the sign of tomorrow's return correctly on 53.1% of the days, more than a coin: it has skill.
- (d) For the VaR 99% use quantile regression at tau = 0.99 on the next-day return.
- (e) The lasso removes the features whose t-statistics are not significant.
- (f) QLIKE punishes an under-forecast of the variance more than an over-forecast by the same factor.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

Points to Remember

- The expected error is bias^2 + variance + noise: flexibility is not free
- Time series: walk-forward with purging; never shuffle days with overlapping targets
- Volatility: linear and penalised models are hard to beat; trees on the HAR features alone lose
- The sign of daily returns: compare with "always up"; no model beats it here
- VaR 1%: a linear quantile regression passes the backtests; a flexible quantile model at 1% does not
- An AI answer is a draft: check the validation scheme, the baseline and the VaR convention
- Next
  - Lecture 13 develops these topics, adds neural networks and data snooping
  - C1 can grow into a team project: BVB stocks, external markets, several horizons
  - Reading: James et al. (2023), Ch. 2, 5, 6 and 8; Franke, Härdle & Hafner (2019), Ch. 19

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